

Machine Learning for Quantum Phase Transitions

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ABSTRACT

Quantum many-body systems exhibit rich emergent behavior, with quantum phase transitions (QPTs) providing key insights into strongly correlated matter. The Bose-Hubbard model serves as a paradigmatic framework to study such transitions, capturing the interplay between kinetic energy and on-site interactions that give rise to the superfluid-Mott insulator transition [1]. However, accurate characterization of QPTs becomes computationally demanding as system sizes grow, motivating the integration of machine learning (ML) techniques into quantum many-body research.

In this work, we explore the use of neural networks to classify quantum phases and detect critical points in the Bose-Hubbard model. By training supervised and unsupervised learning models on observables such as superfluid order parameter, and local occupation number we demonstrate that neural architectures can learn nontrivial order parameters and capture universal features of QPTs without prior knowledge of the underlying physics. This study highlights the potential of approaches like machine learning (ML), pointing towards data-driven strategies for uncovering novel phases of matter and advancing the theoretical understanding of quantum criticality [3-4].

We choose the standard Bose-Hubbard model for implementing and testing of our machine learning models. There are several reasons for choosing this specific model, namely: (i) the model is simple to implement computationally and has been studied well for decades, (ii) the theoretical findings for the model have been verified experimentally, (iii) although simple, it provides a very good fundamental understanding of quantum phase transitions, (iv) the model is straightforward to extend to include higher

order complexities and realize exotic phases like topological insulators, supersolids, etc.

On the ML side we choose to work with neural networks and its variants. Although we have tested other methods, e.g. regression models, in the initial testing itself we found that neural networks easily outperform these aforementioned machine learning methods.

In our studies on the BHM using single site mean-field theory along with ML methods we find that although methods like Multi Layer Perceptron can perform well out-of-the-box (~98% accuracy), they are moderately reliable. It is upon the inclusion of Physics Informed Neural Networks we find that: (i) accuracy improves (~99.3%) as compared to MLP, and (ii) the size of training data can be reduced considerably while maintaining reasonable accuracy.

From our findings we conclude that such advanced ML approaches can be combined with more complex versions of MF method, such as, cluster MF theory, which can improve accuracies further while cutting down the computational effort.

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